

Gunakarthik Naidu Lanka

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PROFESSIONAL SUMMARY

Software Engineer (MSCS 2025) with a 4.0 GPA and 4+ years of Software development practice. Specialized in architecting **distributed systems**, **autonomous agentic workflows**, and **high-performance backend microservices**. Expert in Python, Java, and TypeScript, with a proven track record of reducing system latency by 45% and automating complex manual processes for enterprise-scale platforms. Driven by a "Show > Tell" philosophy and a deep commitment to engineering excellence and data sovereignty.

TECHNICAL SKILLS

Languages: Python, Java (Spring Boot), TypeScript, Node.js, SQL (Postgres/MySQL), C++, Go, HTML/CSS
AI/Agentic: OpenClaw, Playwright, LangGraph, RAG Pipelines, Ollama, Model Quantization (GGUF), Prompt Engineering
Infrastructure: AWS (EC2, S3, Lambda, Fargate), Docker, Kubernetes, Terraform (IaC), CI/CD, GitHub Actions, Linux
Web/Backend: React, Next.js, FastAPI, Hono, Drizzle ORM, WebSockets, Redis, RabbitMQ, REST APIs, GraphQL
Concepts: System Design, Distributed Systems, Microservices, SDLC, Performance Optimization, Security Compliance

PROFESSIONAL EXPERIENCE

Software Engineering Intern, Velocity Tech (Zinio TalentHub) Jan 2024 – Dec 2024

- Engineered high-concurrency backend microservices using **FastAPI** and **Spring Boot**, supporting a matching engine for 5,000+ monthly active users with 99.9% system uptime.
- Architected a high-fidelity NLP parsing engine that transformed unstructured data into searchable JSON, improving extraction accuracy by 30% and reducing manual overhead by 70%.
- Optimized **PostgreSQL** query performance through advanced indexing and schema refactoring, resulting in a 45% reduction in p99 data retrieval latency across high-cardinality tables.
- Streamlined the full SDLC by automating deployment pipelines via **GitHub Actions**, accelerating feature shipping cycles by 25% while maintaining rigorous QA standards.

Full Stack Intern, EPICS at ASU Jan 2022 – May 2022

- Developed end-to-end features for a state-wide resource directory using **React.js** and Python REST APIs, facilitating real-time matching for critical community services.
- Orchestrated containerized deployments via **Docker** and managed AWS cloud infrastructure, ensuring scalable, secure delivery for sensitive healthcare data.
- Established an automated unit-testing suite and led QA cycles, identifying and resolving 20+ production-blocking bugs prior to successful deployment.

PROJECTS

HireAgent (Autonomous Browser Agent) TypeScript, Playwright, PostgreSQL, Ollama

- Architected a self-correcting 6-stage **agentic state machine** to automate multi-step web workflows; achieved 98% task completion rate through a resilient error-recovery layer and exponential backoff strategies.
- Implemented **DOM distillation** to strip 90% of non-essential HTML, enabling local-first 7B LLM inference to navigate complex UIs with 95% accuracy at zero operational cost.

AutoAudit AI (Local-First Inference Engine) C++, GGUF Quantization, Go, Docker

- Designed a "Zero-Cloud" vehicle auditing tool using locally hosted models (DeepSeek-R1/Gemma 3) to ensure 100% data privacy for sensitive industrial records.
- Applied Q4/Q5 GGUF quantization on Apple Silicon (M1 Pro), optimizing token-generation and enabling complex reasoning on 16GB VRAM.
- Engineered a custom KV-cache management system to reduce time-to-first-token (TTFT) by 30%, enabling near-instantaneous reasoning for real-time industrial sensor data audits.

FosterArizona.org (Distributed Platform) Next.js, WebSocket, Redis, AWS

- Engineered a real-time data sync layer using **WebSockets** and **Redis**, providing sub-second availability updates across a distributed network of state-level resource databases.

EDUCATION

Arizona State University, Tempe, AZ	GPA: 4.0/4.0
Master of Science in Computer Science	Dec 2025
Bachelor of Science in Computer Science	Dec 2024